

does copd exacerbation correlate/cause acute myocardial infarction or acute coronary syndrome?

Yes, COPD exacerbations strongly correlate with and likely cause acute myocardial infarction and acute coronary syndrome.

Does COPD exacerbation correlate with or cause acute myocardial infarction or acute coronary syndrome?

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

Multiple large self-controlled case series, meta-analyses, and multinational cohort studies consistently demonstrate a marked, temporally clustered increase in acute coronary events following COPD exacerbations. The evidence supports both correlation and a likely causal contribution, though distinguishing true plaque-rupture events from oxygen supply-demand mismatch (type 2 MI) remains a challenge.

Evidence for Exacerbation-Triggered Cardiac Events

Evidence Strength

Claim



Strong

Acute exacerbations of COPD are associated with a **2.3- to 14.5-fold increased risk** of myocardial infarction or acute coronary syndrome within the first 1–14 days, with risk remaining elevated for up to one year. (Donaldson et al., 2010; Rothnie et al., 2018; Graul et al., 2023; Kunisaki et al., 2018; Reilev et al., 2019)



Strong

Severe (hospitalized) exacerbations confer **greater cardiac risk** than moderate exacerbations, with incidence rate ratios of 2.58 vs. 1.58 for MI. (Rothnie et al., 2018; Kunisaki et al., 2018; Graul et al., 2023; Reilev et al., 2019)



Moderate

Frequent and severe exacerbation history is associated with **long-term elevated MI risk**, with subdistribution hazard ratios reaching 1.82 for two or more severe events. (Wallström et al., 2024; Wallström et al., 2023)



Moderate

The association is partly attributable to **reverse causation and type 2 MI** from hypoxia and tachycardia, which are difficult to distinguish from type 1 plaque-rupture events. (Reilev et al., 2019; Goedemans et al., 2020; Wallström et al., 2024)

FIGURE 1 Evidence strength for key claims linking COPD exacerbations to cardiac events

Temporal Risk and Severity Gradient

The landmark self-controlled case series by Donaldson et al. (2010), analyzing 25,857 COPD patients, established a **2.27-fold increased MI risk** in the 1–5 days immediately following an exacerbation, with risk diminishing progressively thereafter (Donaldson et al., 2010). This temporal clustering was replicated and extended by Rothnie et al. (2018), who found MI risk elevated for 91 days post-exacerbation, with severe exacerbations producing an IRR of 2.58 versus 1.58 for moderate events (Rothnie et al., 2018). The SUMMIT trial post hoc analysis confirmed a **3.8-fold hazard** for composite cardiovascular events within 30 days, rising to 9.9-fold after hospitalized exacerbations (Kunisaki et al., 2018).

A 2023 multinational cohort study across four countries found acute coronary syndrome hazard ratios ranging from 5.59 to 24.57 in the first 7 days post-exacerbation, remaining elevated for 30 days (Hawkins et al., 2023). Graul et al. (2023) reported a composite cardiovascular aHR of 3.19 in the first 14 days, with severe exacerbations driving an aHR of 14.5 (Graul et al., 2023). The 2025 meta-analysis by Pirera et al. confirmed significantly increased ACS risk peaking at 1–30 days and persisting beyond one year (Pirera et al., 2025).

Mechanisms, Limitations, and Clinical Implications

- **Systemic inflammation** during exacerbations—marked by surges in IL-6, TNF- α , CRP, and fibrinogen—drives atherosclerotic plaque instability and thrombosis (Marriott et al., 2024; Goedemans et al., 2020; Siozos et al., 2026).
- **Hypoxemia and tachycardia** during exacerbations create coronary oxygen supply-demand mismatch, precipitating type 2 MIs that may be overrepresented in COPD populations (Goedemans et al., 2020; Reilev et al., 2019; Wallström et al., 2024).
- **Reverse causation** is a plausible alternative explanation for the shortest-latency events, as early MI symptoms may be misattributed to COPD exacerbation and treated as such before cardiac diagnosis (Reilev et al., 2019; De Molina et al., 2018).
- However, the **parallel stroke association** following exacerbations supports a genuine vascular trigger, since strokes are unlikely to be confused with COPD symptoms (Reilev et al., 2019; Donaldson et al., 2010).
- Exacerbation frequency and severity independently predict long-term MI risk, with **two or more severe exacerbations** yielding a subdistribution hazard ratio of 1.82 (Wallström et al., 2024).
- Among patients hospitalized for severe AECOPD, **18.75% experienced an acute cardiovascular event** during hospitalization, with advanced age and lower FEV1 as independent predictors (Siozos et al., 2026).

The consistency of the temporal signal across self-controlled designs—which eliminate time-invariant confounding by smoking, age, and comorbidities—argues for a genuine causal contribution of exacerbations to acute coronary events, though the relative contribution of type 1 versus type 2 MI mechanisms requires further delineation. Preventing exacerbations may decrease the risk of subsequent acute cardiovascular events (Müllerová et al., 2022; Graul et al., 2023).

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